

Week 14 Lab Tasks: Unsupervised Learning (K-Means & Hierarchical Clustering)

Submission Guidelines:

1. Each task must properly show the output. DO NOT Clear the outputs after running.
2. **Download the .ipynb file** from Jupyter Notebook.
3. **Upload the .ipynb** to GCR.
4. **Failure to follow the submission guidelines will result in a deduction of 50% of the obtained marks.**

Task 1: Centroid-Based Customer Segmentation using K-Means Clustering

Problem Statement

Construct a centroid-driven unsupervised clustering model using the **K-Means algorithm** to identify latent customer purchasing segments.

Dataset Details

Use **Mall Customers Dataset**

Expected Attributes:

Attribute Name	Type	Description
CustomerID	Integer	Unique customer identifier
Gender	Categorical	Male / Female
Age	Numeric	Customer age in years
Annual Income (k\$)	Numeric	Annual income in thousand dollars
Spending Score (1-100)	Numeric	Purchasing behavior index

Recommended Features for Clustering:

- Age
- Annual Income (k\$)
- Spending Score (1-100)

Dataset Size:

- Approximately 200 observations
- 5 columns

Requirements

1. Import dataset into DataFrame.
2. Perform data quality auditing.
3. Extract clustering feature matrix.
4. Apply feature standardization.
5. Determine optimal K using Elbow Method and Silhouette Score.
6. Train K-Means model.
7. Generate cluster labels and centroid vectors.
8. Interpret each segment behaviorally.

Mandatory Visualizations

- Elbow curve
- Silhouette score graph
- Cluster scatter plot with centroids
- Pairwise cluster visualization

Task 2: Hierarchical Customer Taxonomy using Agglomerative Clustering

Problem Statement

Develop a hierarchical cluster topology through **Agglomerative Clustering** for customer behavioral grouping.

Dataset Details

Use the **same Mall Customers Dataset** from Task 1.

Required Features:

- Age
- Annual Income (k\$)
- Spending Score (1-100)

Preprocessing Constraint:

Use identical standardized feature matrix from Task 1 to ensure methodological consistency.

Requirements

1. Generate linkage matrix using Ward criterion.
2. Construct dendrogram.
3. Determine optimal number of clusters from merge distances.
4. Train AgglomerativeClustering model.
5. Compare obtained clusters with K-Means.

Mandatory Visualizations

- Dendrogram
- Scatter plot of hierarchical clusters
- Comparative plot (K-Means vs Hierarchical)

Task 3: Mixed-Type Attribute Clustering using Encoded Categorical Features

Problem Statement

Evaluate clustering behavior after incorporating categorical attributes into feature space.

Dataset Details

Use **Mall Customers Dataset**

Features to Use:

Numerical:

- Age
- Annual Income (k\$)
- Spending Score (1-100)

Categorical:

- Gender

Encoding Requirement:

Transform Gender using:

- One-Hot Encoding

Requirements

1. Encode Gender.
2. Merge encoded vectors with scaled numerical attributes.

3. Apply K-Means clustering.
4. Compare results with Task 1 numeric-only clustering.

Mandatory Visualizations

- Correlation heatmap
- Cluster plots before/after encoding
- Silhouette comparison chart

Task 4: Influence of Feature Scaling on Distance-Based Clustering

Problem Statement

Empirically investigate the effect of feature magnitude disparities on clustering outcomes.

Dataset Details

Use **Mall Customers Dataset**

Features:

- Age
- Annual Income (k\$)
- Spending Score (1-100)

Experimental Variants:

1. Raw (Unscaled) dataset
2. Standardized dataset

Requirements

1. Apply K-Means on raw and scaled data.
2. Apply Hierarchical clustering on raw and scaled data.
3. Compare cluster memberships and quality metrics.

Mandatory Visualizations

- Raw vs scaled cluster plots
- Inertia comparison graph
- Silhouette comparison graph

Task 5: Outlier-Resilient Cluster Diagnostics

Problem Statement

Examine the impact of anomalous tuples on centroid displacement and hierarchical merge structure.

Dataset Details

Use **Mall Customers Dataset**

Features:

- Age
- Annual Income (k\$)
- Spending Score (1-100)

Additional Instruction:

Artificially inject 3–5 extreme-value records **or** detect natural outliers.

Requirements

1. Detect outliers using IQR.
2. Run K-Means before and after removal.
3. Compare centroid coordinates.
4. Run Hierarchical clustering before and after cleaning.

Mandatory Visualizations

- Boxplots
- Pre/post outlier cluster plots
- Centroid shift visualization

Task 6: Multi-Domain Clustering on Alternative Dataset

Problem Statement

Apply both algorithms on a second dataset to validate generalization.

Dataset Details

Use **Iris Dataset** from Scikit-Learn.

Attributes:

Feature	Type
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Sepal Length	Numeric
Sepal Width	Numeric
Petal Length	Numeric
Petal Width	Numeric

Dataset Size:

- 150 records
- 4 numerical features
- 3 natural species classes (for reference only)

Requirements

1. Load Iris dataset using sklearn.
2. Ignore labels during training.
3. Apply K-Means and Hierarchical clustering.
4. Compare generated clusters with actual species labels.

Mandatory Visualizations

- PCA 2D projection of clusters
- Confusion matrix (true labels vs cluster IDs after mapping)
- Dendrogram

Important Note

- Using labels to choose clusters during training would violate pure unsupervised methodology.
- Using labels **only for post-analysis** is completely standard practice.